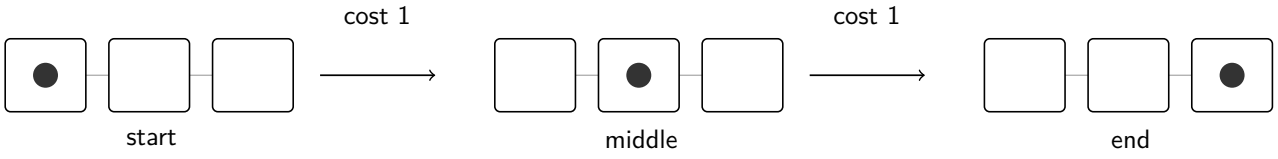
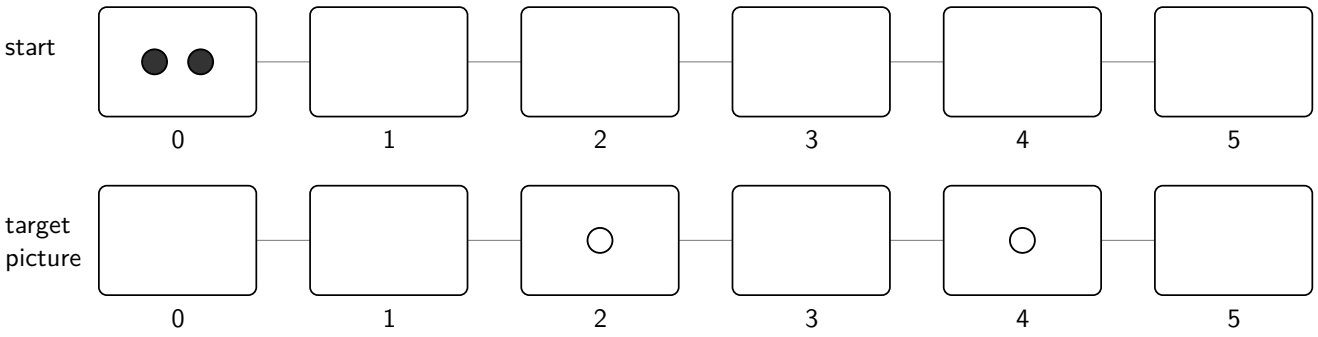
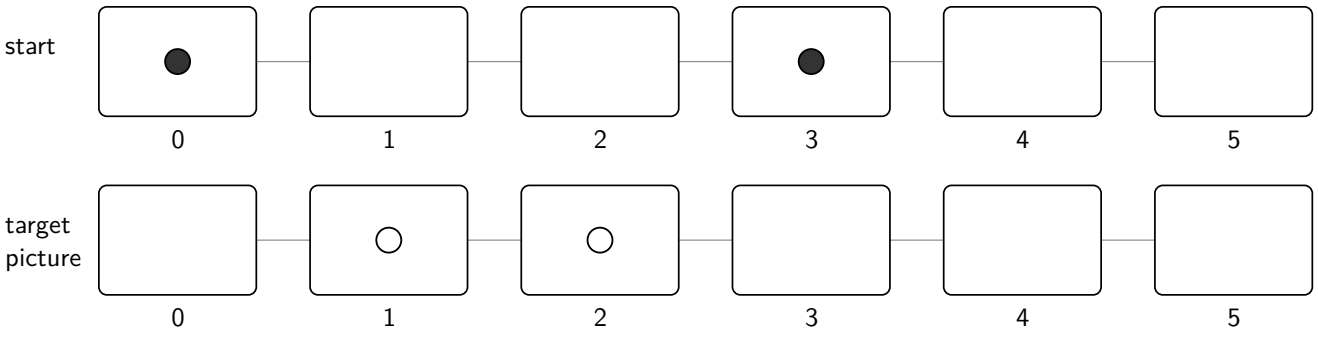
Move one counter at a time to a neighboring bin. Each crossing costs 1. Bins hold any number of counters, and counters may pass through occupied bins. Any counter may fill any target place. Only start counters move; the target picture stays fixed. Write each move as you make it.

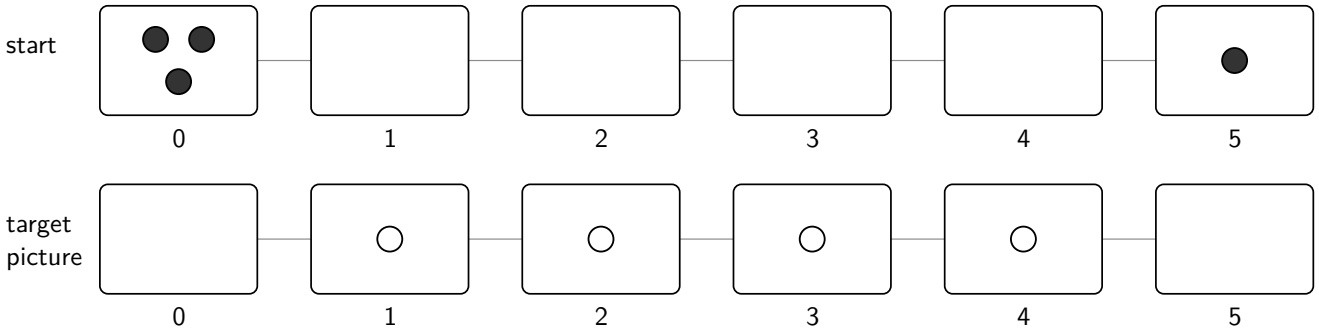


Problem 1: Match each target using the least work you can.

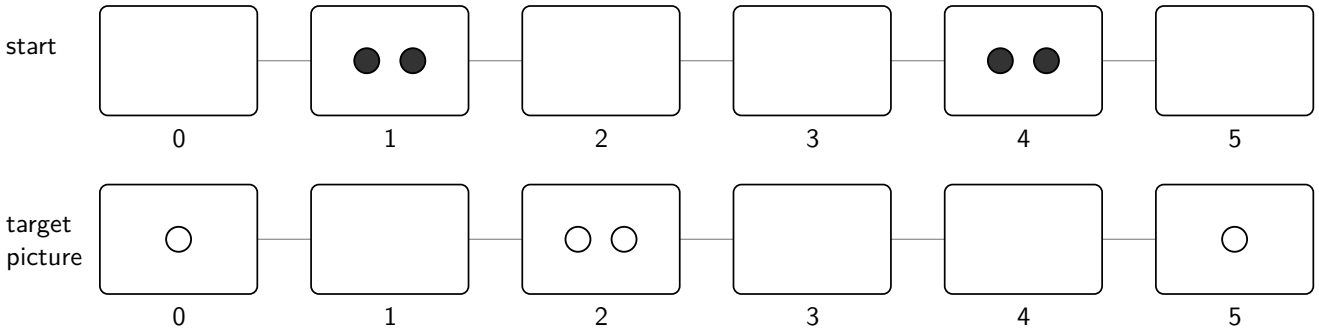




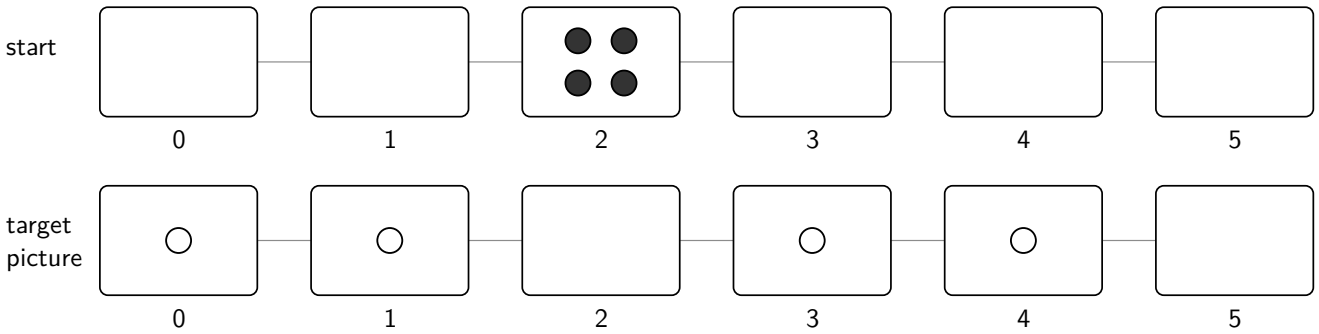
Problem 2: Make each start match its target. How little work can you use?



work: _____

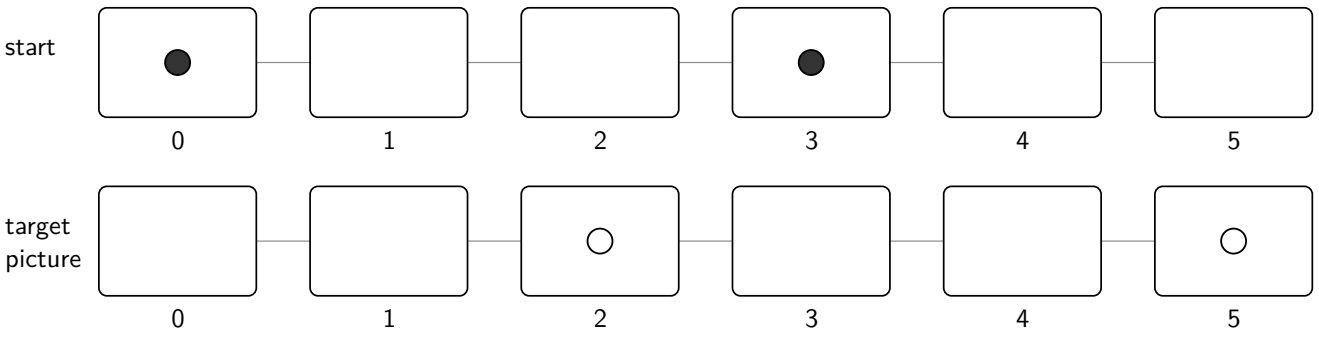


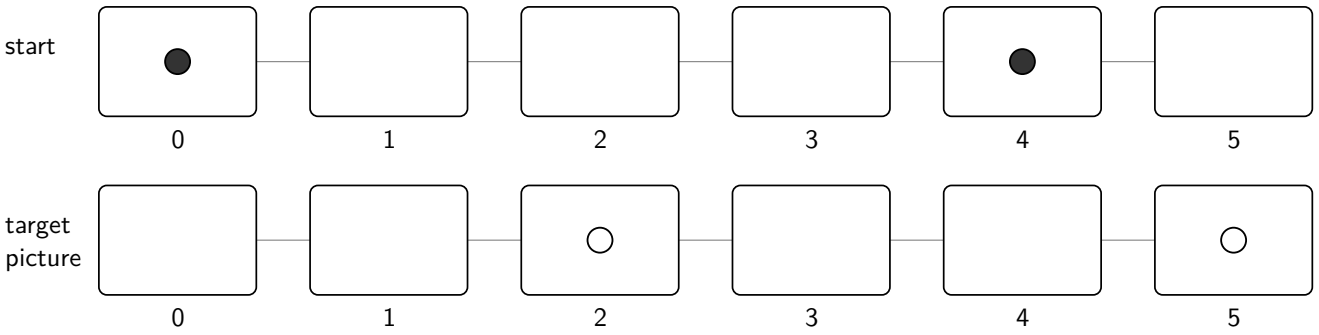
work: _____



work: _____

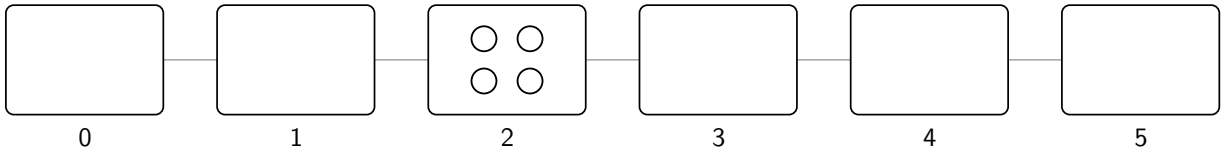
Problem 3: A delivery carries one counter straight to an unfilled target place. Once delivered, that counter stays there. Try this rule: always make the shortest remaining delivery. For each pair, can another plan use less work?



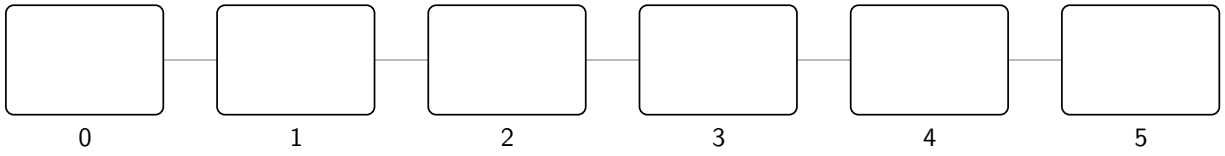


Problem 4: Use four counters and keep this target fixed. Find different starts whose least work is 4. Then find starts whose least work is 7.

target picture

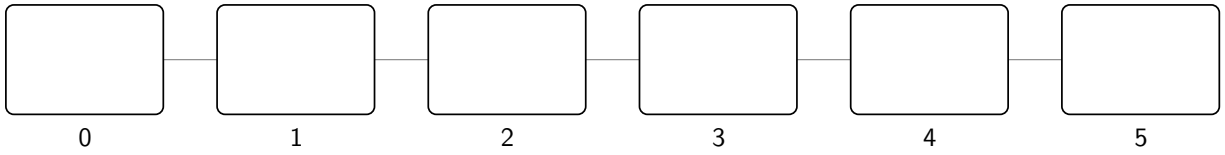


start



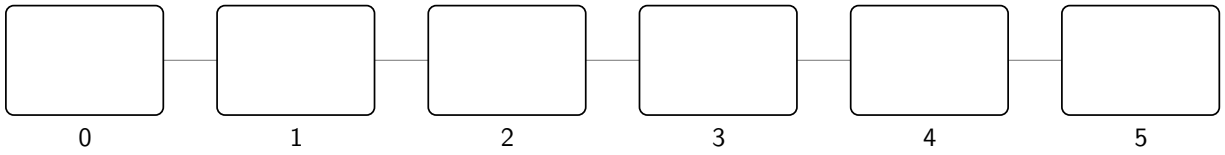
least work: _____

start



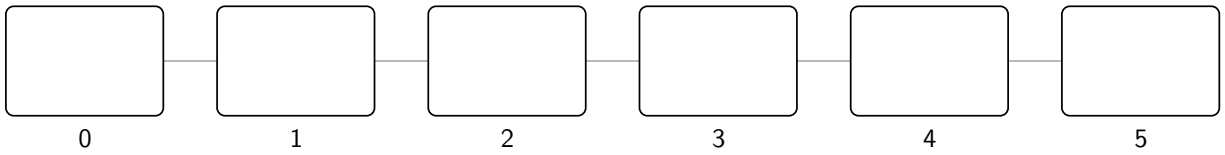
least work: _____

start



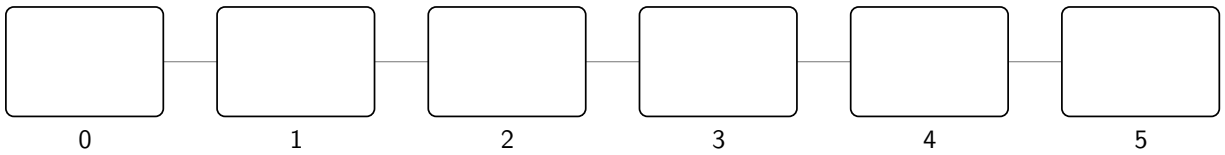
least work: _____

start



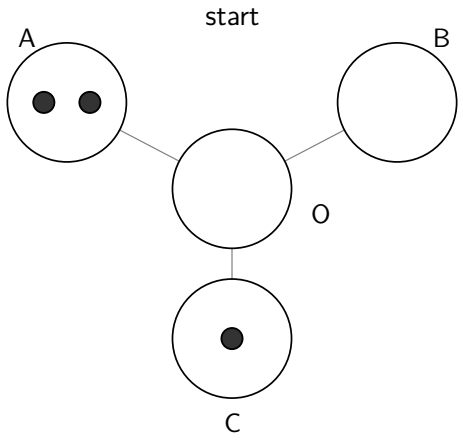
least work: _____

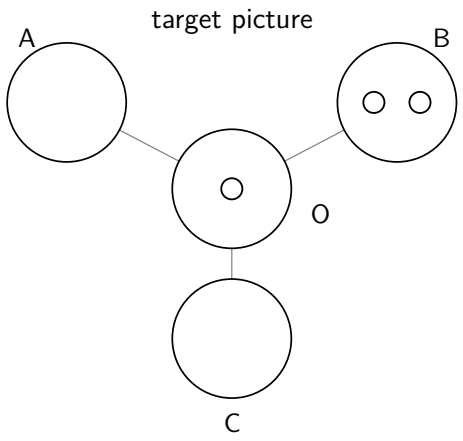
start



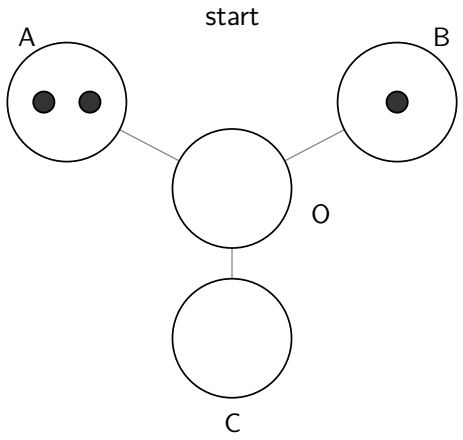
least work: _____

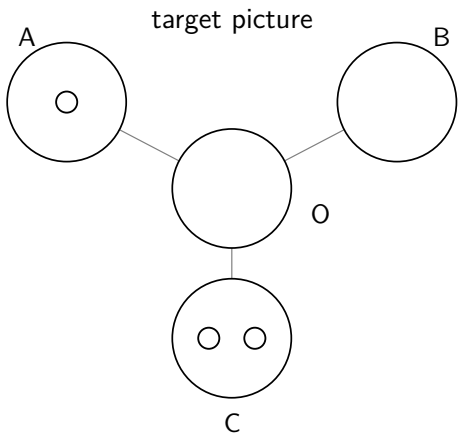
Problem 5: Move counters along the roads to match each target. One counter crossing one road costs 1. Find the least work for each pair. Then draw a new start and target with three counters each whose least work is 6.



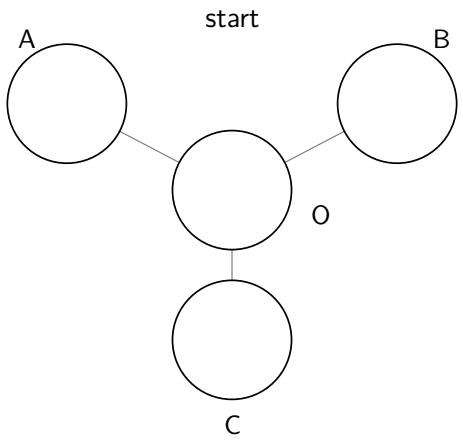


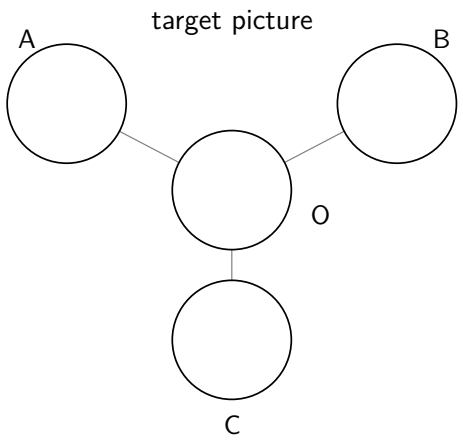
work: _____





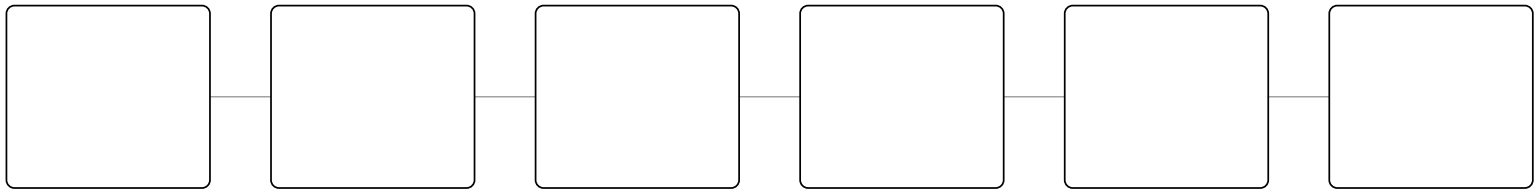
work: _____



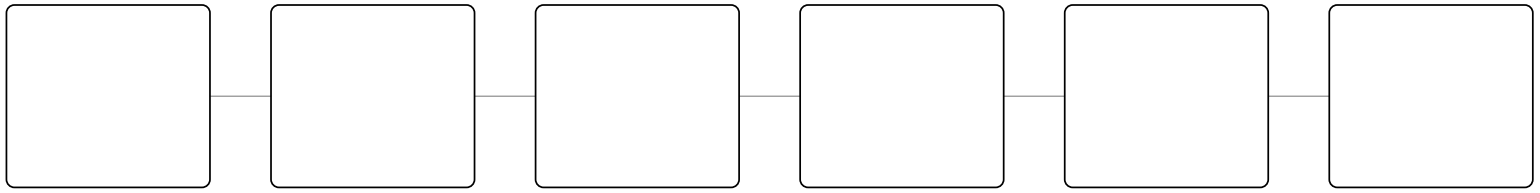


Problem 6: One player draws a target with four counters. The other puts four counters on the start row so the least work is exactly 5. Check it, then swap. Can a target make this impossible?

start



target picture



work: _____